Inverse Eigenvalue Problem for Persymmetric Matrices with Prescribed Pattern

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1 Definition and first properties

Let J_n be the $n \times n$ reversal matrix: its (i, j) entry is 1 when $i + j = n + 1$ and 0 otherwise. Thus $J_n^T = J_n$ and $J_n^2 = I_n$. A real matrix $A = (a_{ij})$ is *persymmetric* when reflection across its anti-diagonal leaves its entries unchanged. In indices, this means

$$a_{ij} = a_{n+1-j,n+1-i}, \qquad 1 \le i, j \le n. \tag{1}$$

Equivalently, $A = J_n A^T J_n$. Multiplying this identity by J_n on the left shows that $J_n A$ is an ordinary symmetric matrix. Conversely, every symmetric matrix S gives a persymmetric matrix $J_n S$. Hence the space of real persymmetric $n \times n$ matrices has dimension $n(n + 1)/2$.

Example 1. *For $n = 3$, the matrix from the project brief [2]*

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 2 \\ 6 & 4 & 1 \end{pmatrix}$$

is persymmetric: the entries a_{11} and a_{33} agree, as do a_{12} and a_{23}, and a_{21} and a_{32}. The three entries on the anti-diagonal are fixed by the reflection and need not equal one another. This example is not symmetric, since $a_{12} \neq a_{21}$.

Persymmetry alone does not force real eigenvalues. The 2×2 analysis below supplies an explicit counterexample. A *bisymmetric* matrix has the additional requirement $A = A^T$.

The first two lemmas of Julio and Soto [1] describe operations that preserve these two kinds of symmetry. We give short proofs and examples to make the distinction between them explicit. Write $A^F = J_n A^T J_n$ for the flip-transpose of A. In particular, $(AB)^F = B^F A^F$ whenever the products are defined.

Lemma 1 (Closure of persymmetric matrices)**.** *If A and B are real persymmetric matrices of the same order, then $sA + tB$ is persymmetric for any $s, t \in \mathbb{R}$. If A is invertible, A^{-1} is persymmetric. The transpose A^T is also persymmetric.*

Proof. The first claim follows from $(sA + tB)^F = sA^F + tB^F = sA + tB$. For the second, the identity $A^{-1}A = I$ gives $A(A^{-1})^F = I$ after flip-transposition, so $(A^{-1})^F = A^{-1}$. Finally, $(A^T)^F = (A^F)^T = A^T$. □

Example 2. *Take*

$$P = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}.$$

Both matrices are persymmetric. So are $P + Q = \left(\begin{smallmatrix} 3 & 5 \\ 3 & 3 \end{smallmatrix}\right)$, $P^{-1} = \left(\begin{smallmatrix} 2 & -1 \\ -3 & 2 \end{smallmatrix}\right)$, and $P^T = \left(\begin{smallmatrix} 2 & 3 \\ 1 & 2 \end{smallmatrix}\right)$. The inverse illustrates that persymmetry can be preserved even when nonnegativity is lost.

Lemma 2 (Closure of bisymmetric matrices)**.** *If A and B are real bisymmetric matrices of the same order, then $sA + tB$ is bisymmetric for any $s, t \in \mathbb{R}$. If A is invertible, A^{-1} is bisymmetric. Moreover, A^T, A^F, and J_nA are bisymmetric, and $J_nA = AJ_n$.*

Proof. Lemma 1 supplies the persymmetry of the linear combination, inverse, and transpose. Their symmetry follows from transposing $sA + tB$ and from $(A^{-1})^T = (A^T)^{-1} = A^{-1}$. Also $A^T = A^F = A$. Since $A = J_nA^TJ_n = J_nAJ_n$, multiplying by J_n on the left yields $J_nA = AJ_n$. Hence $(J_nA)^T = A^TJ_n = AJ_n = J_nA$ and $(J_nA)^F = A^FJ_n^F = AJ_n = J_nA$. □

Example 3. *For*

$$C = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad J_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

we have $J_2C = CJ_2 = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$. Both C and J_2C are bisymmetric. Also $C^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ is bisymmetric.

2 Patterns and the two-dimensional case

We use $\star$ for a required nonzero entry, $\times$ for an entry whose value is unrestricted, and $+$, $-$ and 0 for a positive, negative and zero entry respectively. Any prescribed pattern must respect (1); reflected positions carry the same requirement. For 2×2 matrices, the diagonal positions are paired, whereas each anti-diagonal position is fixed separately. The three sketched families can therefore be written as

$$\mathcal{F}_1 : \begin{pmatrix} \star & \times \\ \times & \star \end{pmatrix}, \qquad \mathcal{F}_2 : \begin{pmatrix} \star & 0 \\ \star & \star \end{pmatrix}, \qquad \mathcal{F}_3 : \begin{pmatrix} + & 0 \\ - & + \end{pmatrix}.$$

In each display, the two diagonal symbols describe the *same* entry value, because persymmetry requires $a_{11} = a_{22}$. The third family is a sign-restricted part of the second. For instance,

$$\begin{pmatrix} 1 & 2 \\ -3 & 1 \end{pmatrix} \in \mathcal{F}_1, \qquad \begin{pmatrix} 2 & 0 \\ 3 & 2 \end{pmatrix} \in \mathcal{F}_2, \qquad \begin{pmatrix} 2 & 0 \\ -3 & 2 \end{pmatrix} \in \mathcal{F}_3.$$

Every real persymmetric 2×2 matrix has the form

$$A_2 = \begin{pmatrix} a & b \\ c & a \end{pmatrix}.$$

Writing the characteristic polynomial in the monic convention gives

$$\chi_{A_2}(x) = \det(xI_2 - A_2) = (x - a)^2 - bc, \qquad \lambda_{1,2} = a \pm \sqrt{bc}. \tag{2}$$

The product bc, rather than the signs of b and c separately, determines the spectral case:

- If $bc > 0$, there are two distinct real eigenvalues. For example, $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has spectrum $\{0, 2\}$.
- If $bc = 0$, a is a repeated eigenvalue. For example, $\left(\begin{smallmatrix} 2 & 0 \\ -3 & 2 \end{smallmatrix}\right)$ has characteristic polynomial $(x - 2)^2$. It is not diagonalizable, since the off-diagonal entry is nonzero.
- If $bc < 0$, the eigenvalues form a nonreal conjugate pair. For example, $\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ has eigenvalues $1 \pm i$.

Thus the locations of nonzero entries alone do not determine the spectral case: the signs of b and c determine the sign of bc.

3 The three-dimensional nilpotent question

The general real persymmetric 3×3 matrix is

$$A_3 = \begin{pmatrix} a & b & d \\ c & e & b \\ f & c & a \end{pmatrix}. \tag{3}$$

Expanding $\det(xI_3 - A_3)$ yields

$$\begin{aligned} \chi_{A_3}(x) &= x^3 - (2a+e)x^2 + (a^2 + 2ae - 2bc - df)x - \det A_3, \\ \det A_3 &= a^2 e - 2abc + b^2 f + c^2 d - def. \end{aligned} \tag{4}$$

Consequently, $\chi_{A_3}(x) = x^3$ exactly when

$$e = -2a, \qquad 3a^2 + 2bc + df = 0, \qquad -2a^3 - 2abc + b^2 f + c^2 d + 2adf = 0. \tag{5}$$

These are three coefficient equations, not just the trace condition $e = -2a$. By the Cayley–Hamilton theorem they are also equivalent to $A_3^3 = 0$.

Example 4. *The upper shift*

$$U = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

is persymmetric and has $\chi_U(x) = x^3$. Here $U^2 \neq 0$ but $U^3 = 0$. This is a nonnegative example.

Example 5. *The signed matrix*

$$N = \begin{pmatrix} 0 & 1 & 1 \\ 2 & 0 & 1 \\ -4 & 2 & 0 \end{pmatrix}$$

is also persymmetric and satisfies $\chi_N(x) = x^3$. Positive and negative products cancel in the coefficients of its characteristic polynomial. Direct multiplication gives $N^2 \neq 0$ and $N^3 = 0$. This illustrates why the signed-pattern problem differs from the nonnegative problem.

The cancellation is visible algebraically. When $a = e = 0$, the coefficient of x is $-(2bc+df)$, while the determinant is $b^2 f + c^2 d$. For N, both expressions vanish because $2(1)(2)+(1)(-4) = 0$ and $(1)^2(-4) + (2)^2(1) = 0$. The signs of the entries make these cancellations possible.

If A_3 is entrywise nonnegative and $\chi_{A_3}(x) = x^3$, the first equation in (5) forces $a = e = 0$: a sum of nonnegative diagonal entries can vanish only when each one vanishes. The signed example above does not satisfy entrywise nonnegativity. The accompanying Mathematica notebook checks (4), derives (5), and tests these examples symbolically.

4 Rank-one nilpotent persymmetric matrices

We now seek families of real matrices that are simultaneously rank one, nilpotent, and persymmetric. The entries, including the diagonal entries, are unrestricted apart from these three requirements. A rank-one matrix has all its columns proportional to one nonzero column, so it can be written as an outer product

$$A = uv^T, \qquad u, v \in \mathbb{R}^n \setminus \{0\}.$$

Its jth column is $v_j u$. Nilpotency means that $A^k = 0$ for some positive integer k. The outer-product representation reduces this condition to a scalar equation, since

$$A^2 = (uv^T)(uv^T) = u(v^T u)v^T = (v^T u)A.$$

Thus $v^T u = 0$ gives $A^2 = 0$. If $v^T u \neq 0$, every positive power of A is a nonzero scalar multiple of A, so A is not nilpotent.

Lemma 3 (Rank-one nilpotent persymmetric matrices)**.** *A real $n \times n$ matrix A has rank one, is nilpotent, and is persymmetric if and only if*

$$\boxed{A = \lambda J_n v v^T, \qquad \lambda \in \mathbb{R} \setminus \{0\}, \quad v \in \mathbb{R}^n \setminus \{0\}, \quad v^T J_n v = 0.} \tag{6}$$

Every such matrix satisfies $A^2 = 0$ and $A \neq 0$.

Proof. Suppose first that A has all three properties. Write $A = uv^T$ with $u, v \neq 0$. Persymmetry implies

$$uv^T = J_n A^T J_n = (J_n v)(J_n u)^T.$$

The columns on the left span the line generated by u, while those on the right span the line generated by $J_n v$. Since these are the same nonzero matrix, $u = \lambda J_n v$ for some nonzero real scalar λ. Hence $A = \lambda J_n v v^T$. The scalar is retained because a negative multiple cannot be absorbed into vv^T by rescaling a real vector.

Multiplication now gives

$$A^2 = \lambda^2 J_n v (v^T J_n v) v^T = \lambda (v^T J_n v) A.$$

If $v^T J_n v \neq 0$, induction gives $A^k = (\lambda v^T J_n v)^{k-1} A \neq 0$ for every $k \geq 2$, contrary to nilpotency. Therefore $v^T J_n v = 0$.

Conversely, suppose (6) holds. Both $J_n v$ and v are nonzero, so A has rank one. Also

$$J_n A^T J_n = J_n (\lambda v v^T J_n) J_n = \lambda J_n v v^T = A,$$

so A is persymmetric. Finally, $A^2 = \lambda (v^T J_n v) A = 0$, proving nilpotency. □

The remaining condition pairs coordinates in reverse order. Explicitly,

$$v^T J_n v = \begin{cases} 2 \displaystyle\sum_{i=1}^{m} v_i v_{n+1-i}, & n = 2m, \\ 2 \displaystyle\sum_{i=1}^{m} v_i v_{n+1-i} + v_{m+1}^2, & n = 2m+1. \end{cases} \tag{7}$$

For example, in even dimension, writing $v = (a^T, (J_m b)^T)^T$ with $a, b \in \mathbb{R}^m$ gives $v^T J_n v = 2a^T b$. Thus perpendicular vectors a and b, not both zero, give the required families. In odd dimension, a middle coordinate z adds z^2, giving the condition $2a^T b + z^2 = 0$.

Complete solution in dimension three

For $v = (p, q, r)^T$, the reversal matrix and the scalar condition are

$$J_3 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad v^T J_3 v = 2pr + q^2 = 0.$$

The corresponding matrices are

$$A = \lambda \begin{pmatrix} pr & qr & r^2 \\ pq & q^2 & qr \\ p^2 & pq & pr \end{pmatrix}, \qquad \lambda \neq 0. \tag{8}$$

The equation $2pr + q^2 = 0$ also shows that the diagonal has the form $(a, -2a, a)$, in agreement with the trace condition in (5).

If $p \neq 0$, the parameters p and q are free, and r is forced:

$$r = -\frac{q^2}{2p}.$$

For instance, $p = 1$ and $q = 2$ give $r = -2$. With $\lambda = 1$, this yields

$$A = \begin{pmatrix} -2 & -4 & 4 \\ 2 & 4 & -4 \\ 1 & 2 & -2 \end{pmatrix}.$$

Its rows are multiples of the nonzero third row, its entries agree under anti-diagonal reflection, and $A^2 = 0$ by Lemma 3.

To complete the solution, the case excluded by division must also be included. If $p = 0$, then $2pr + q^2 = 0$ forces $q = 0$, while $v \neq 0$ requires $r \neq 0$. Consequently, every real rank-one nilpotent persymmetric 3×3 matrix is obtained from (8) using exactly the following two cases:

$$\boxed{\begin{aligned} & v = \begin{pmatrix} p \\ q \\ -q^2/(2p) \end{pmatrix}, \quad p \in \mathbb{R} \setminus \{0\}, \quad q \in \mathbb{R}, \\ & \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \lambda \in \mathbb{R} \setminus \{0\}. \\ \text{or} \quad & v = \begin{pmatrix} 0 \\ 0 \\ r \end{pmatrix}, \qquad\qquad r \in \mathbb{R} \setminus \{0\}, \end{aligned}} \tag{9}$$

References

[1] A. I. Julio and R. L. Soto, "Persymmetric and bisymmetric nonnegative inverse eigenvalue problem," *Linear Algebra and its Applications* **469** (2015), 130–152. doi:10.1016/j.laa.2014.11.025.

[2] H. Šmigoc, "Inverse Eigenvalue Problem for Persymmetric Matrices with Prescribed Pattern," project brief supplied for the final-year project (2026).